

Corrigé — Exercice 20

A. 1) $f'(x) = \frac{1}{2\sqrt{6+x}} > 0$: croissante. **2)** $f(3) = \sqrt{9} = 3$; $f([0, 3]) = [\sqrt{6}, 3] \subset [0, 3]$. **B. 3)** $u_0 = 0 \in [0, 3]$; si $0 \leq u_n \leq 3$, $u_{n+1} = f(u_n) \in [0, 3]$. **4)** $u_{n+1}^2 - u_n^2 = 6 + u_n - u_n^2 = -(u_n - 3)(u_n + 2) \geq 0$: croissante. **5)** croissante majorée par 3 : converge; $\ell = \sqrt{6 + \ell} \Rightarrow \ell^2 - \ell - 6 = 0 \Rightarrow \ell = 3$.

